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South Korea

Seongwoo Park

Electrical Engineer

PROFILE

Graduated from Virginia Tech, majored in electrical engineering with a focus in Controls, Robotics, Autonomy. Highly motivated to do research related in object detection using deep learning, image processing, computer vision.

INTERESTS

- Physical AI
- Computer Vision
- Object Detection
- Image Segmentation
- Humanoid Robot
- Autonomous Robot
- Autonomous Vehicle

LANGUAGE

1. Python
2. C++
3. C
4. C#

Additional Skills

1. LiDAR SLAM
2. Arduino
3. ArUco Markers
4. ROS2
5. YOLO
6. Pytorch

EXPERIENCE

Analyzing Construction Pollution Using Object Detection
Undergraduate Research | *May 2023 - August 2023*

- Trained a model by labeling 500+ images of construction vehicles and had 92.5% of mAP, 93.4% of Precision, and 92.0% of recall
- Collected air quality data using Enviro + Raspberry Pi and analysis indicated a positive correlation between increased vehicle presenece and elevated levels of air pollution

AI-Agent Operated Autonomous Ice Cream Dispensing Machine
XYZ Corporation | *October 2024 - November 2024*

- Implemented face recognition to authenticate customers, seamlessly integrating order tracking and personalized chat history recall.
- Developed AI agent that processes customer order via chat interface, locates the ice cream cup and initiates automated ice cream dispense.

Human Following AI Caddie
XYZ Corporation| *November 2024 - December 2024*

- Developed a human following storage robot using camera and capture images of swing pose.
- Utilized LiDAR sensor to integrat object avoidance algorithm to ensure safe navigation while following human

Mandatory Military Service
Republic of Korea Army| *October 2019 - May 2021*

- Served as a squad leader in the mechanized infantry and learned how to become a leader by serving and helping others during the training sessions.

2019 Summer Conference
Korea Development Institute| *October 2019 - May 2021*

- A two-month long program where I was responsible for hosting foreign diplomats and worked as a staff to lead and guide diplomats.

EDUCATION

Virginia Polytechnic Institute and State University

Digital Image Processing

- Implemented algorithms for noise reduction and image enhancement.
- Designed advanced image analysis methods for compression, segmentation, and representation.

Machine Learning

- Developed supervised classification models using logistic regression and SVMs.
- Conducted in-depth reviews of recent object detection research.

Major Design Experience

- Developed an AR headset application for QR code scanning and valve/pipe detection using YOLO v7 to improve submarine maintenance efficiency.